Mainframe vs. Clustered Cloud-Native Infrastructure: A Parametric TCO Simulation Tool for Enterprise OLTP Workloads

OUTLINE

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- PROJECT AIMS
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- SCOPE
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- DIFFICULTIES ENCOUNTERED
- TASKS TO BE COMPLETED IN THE SECOND SEMESTER

DEFINITION OF THE PROBLEM

What Are We Doing?

- Cost-effectiveness comparison of mainframe based infastructure vs. clustered cloud-native infrastructure under OLTP workloads
- A simulation tool that takes workload and SLA parameters as input
- Minimum-cost feasible architecture recommendation
- The comparison covers the Total Cost of Ownership (TCO)



aws

Aurora

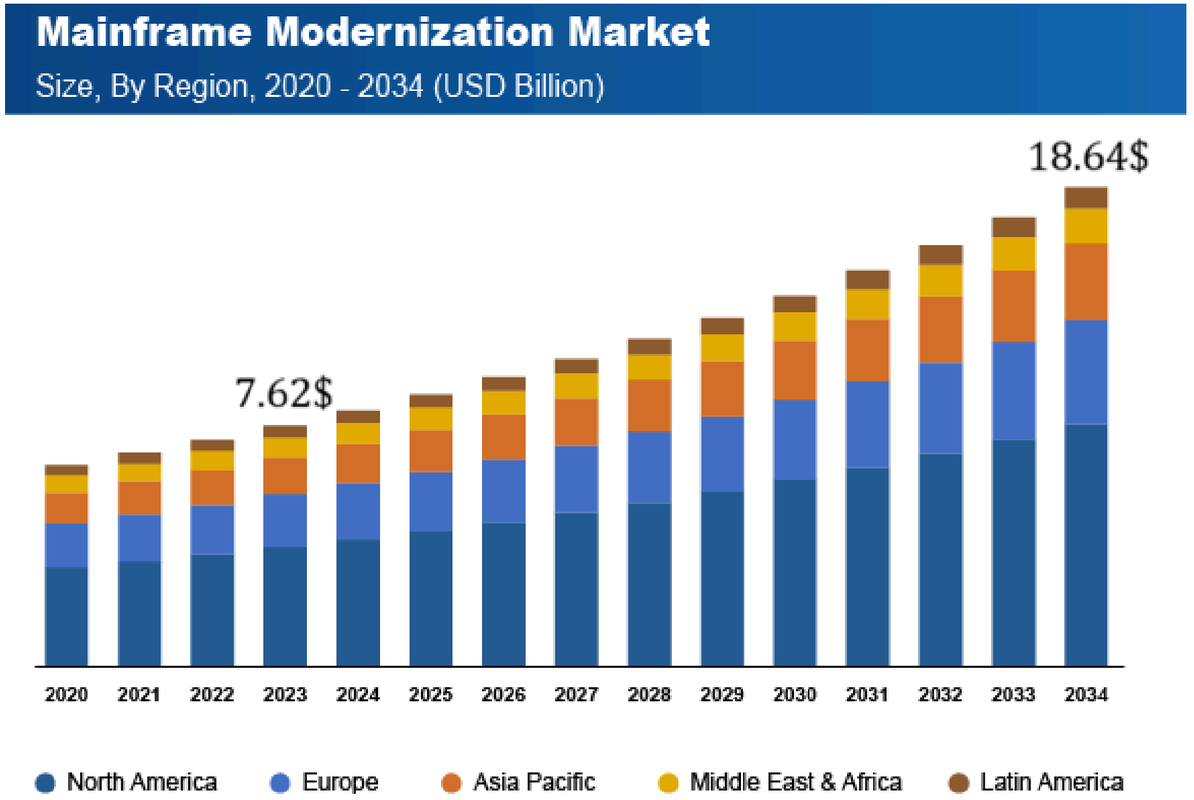
PostgreSQL

DEFINITION OF THE PROBLEM

Why Is the Problem Important?

- %70 of Fortune 500 companies use Mainframes
- An Accenture survey of 150 banking IT executives found that

  4 in 5 plan to migrate mainframe workloads to the cloud



"The cloud-based modernization segment held the largest revenue share in 2024, due to its ability to offer cost efficiency and agility."
www.polarismarketresearch.com

DEFINITION OF THE PROBLEM

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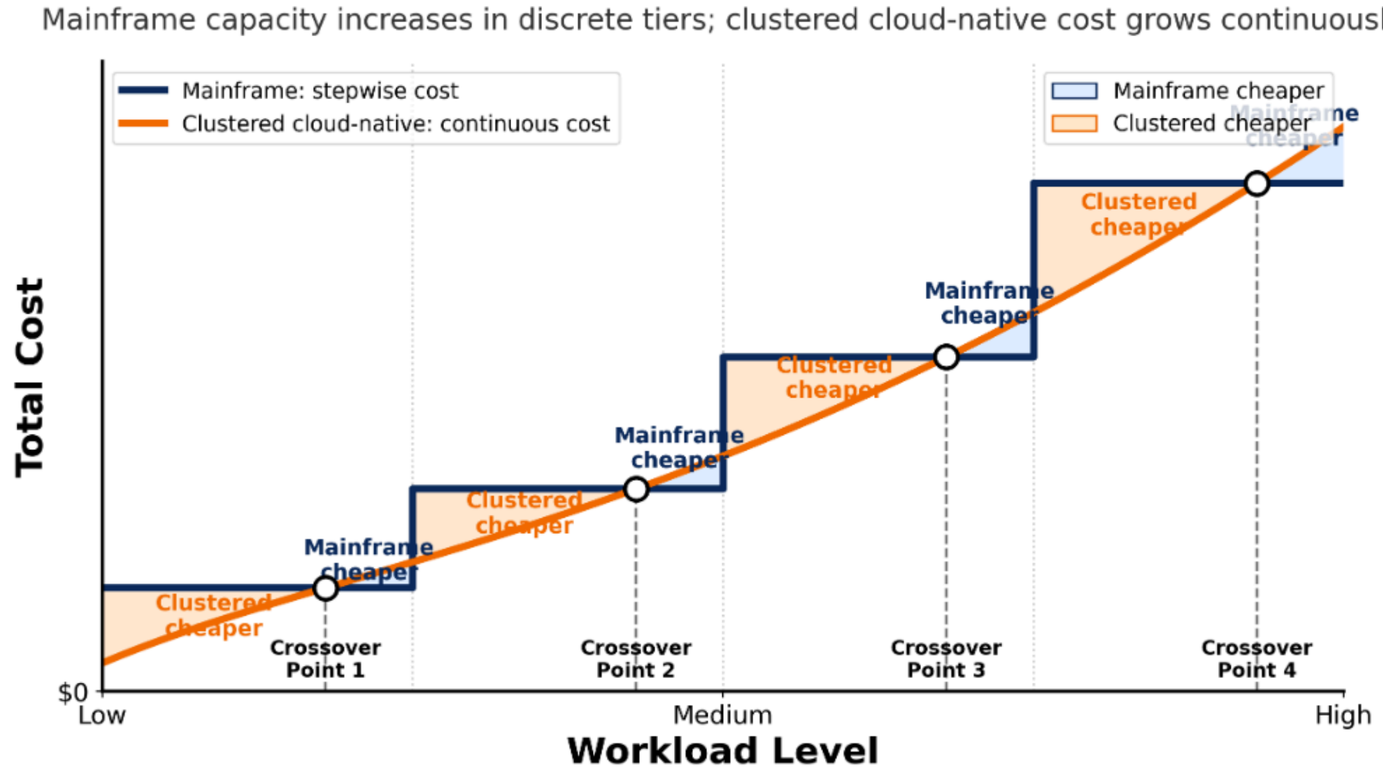
- An analysis of 29 mainframe-to-cloud migrations (2020-2025, $3M-$45M budgets) found 66% failed to meet stated objectives
  - 31% abandoned mid-project
  - **24% completed but reverted to mainframe within 12 months**
  - 11% ongoing “zombie” projects
  - Budget overrun: +287% of original estimate
  - Timeline extension: +22.4 months beyond planned completion

The right infrastructure decision, made before commitment, is worth more than any recovery plan

DEFINITION OF THE PROBLEM

What Is the Motivation for This Project?

- Preventing costs induced by wrong architecture choices.
- **The project's core insight**: Mainframe capacity scales in large discrete steps, while clustered scales continuously, creating predictable cost crossover points depending on workload level.



AIMS OF THE PROJECT

The model shall be built upon an underlying cost and feasibility model that bridges vendor documentation, industry benchmarks, and real-world migration patterns to ensure both sides of the comparison are modeled with the same discipline and at the same level of depth.

01 Build capacity-throughput bridges from vendor documentation and real-world benchmarks to ground the model in realistic, calibrated parameters

02 Simulate annual workload profiles to surface cost risks and capacity bottlenecks over time

03 Reveal the crossover points where each paradigm becomes cheaper depending on workload level

04 Visualize how each architecture meets the institution's requirements side by side, enabling direct comparison

RELATED WORK

CloudSim

- Cloud data center and VM provisioning simulation.
- Mainly used for resource allocation and scheduling studies.
- Not designed for mainframe MSU/CoD billing or OLTP TCO comparison.

OMNeT++

- Discrete-event simulation framework.
- Primarily used for building network and distributed-system simulators.
- Not focused on yearly infrastructure TCO or enterprise OLTP cost modeling.

Novelity

- Domain-specific, SLA-aware TCO model for enterprise OLTP infrastructure
- Workload sweep to reveal cost crossover points

Layer	Mainframe side	Clustered side((AWS cloud-native)
Compute + OS	IBM z16 + z/OS	EKS managed control plane
Transaction / DB	CICS + Db2 for z/OS	Aurora PostgreSQL
Messaging	IBM MQ	Amazon MQ / SQS
Network	OSA-Express and FICON SAN	AWS Backup + cross-region
Storage	DS8900F block storage	EBS gp3/io2 + S3
Backup / DR	GDPS	AWS Backup + cross-region
Operations	Sysprog + DBA	Cloud engineer + SRE

SCOPE

Workload focus: OLTP only. This is the domain where the mainframe vs. clustered question is open. Both architectures are used in production

Architecture scope: Bottom layer of enterprise IT. We model the full infrastructure stack that runs and supports transactional workloads.

Key assumptions

- OLTP I/O characteristics held constant inside the model.
- IBM hardware and software list prices are estimated from independent industry sources and sensitivity-tested.
- Availability is evaluated conservatively against independent survey measurements (ITIC).
- Planned maintenance is accounted for via an industry-observed rule: only ~10% of total observed downtime
- Clustered throughput scaling sublinearity modeled from benchmark data

What is explicitly out of scope?

- Application-layer modernization
- OLAP, analytics, ML training, batch-only workloads
- Network infrastructure below the storage layer
- Licensing negotiation dynamics

METHODOLOGY AND TECHNICAL APPROACH

Overall Approach

For 8 user inputs, both mainframe and clustered configuration spaces are searched to find the cheapest feasible option.

Parameter	Description	Unit / Typical Value Range
throughput	Number of transactions per second	TPS (e.g., 10,000–500,000)
rni	Transaction complexity (CPU + IOPS intensity)	low / average / high
latency_ms + percentile	Target response time + percentile	e.g., p99 < 200 ms
availability	Target uptime	0.999–0.999999
data_volume_tb	Initial data volume	TB (e.g., 10–1000)
data_growth_pct	Annual data growth rate	% (default: 15)
rpo_seconds	Maximum acceptable data loss	seconds (0 = sync)
rto_minutes	Maximum acceptable outage	minutes

END TO END FLOW

Step 1 — A realistic yearly traffic shape is generated at hourly granularity from the user's average TPS and peak characteristics.

Step 2 — Topologies that cannot meet the user's uptime/RPO/RTO targets are eliminated on each side.

Step 3 — For each surviving topology, the cheapest hardware configuration that can handle the workload within latency limits is identified.

Step 4 — Sweeping across input ranges reveals where each architecture becomes the better choice.

TASKS ACCOMPLISHED

1- Mainframe model — compute + availability + yearly simulation

- Full z16 capacity catalog (128 tiers) with calibrated CP-power throughput bridge
- IBM Resiliency 4-tier availability model, dual-bounded (ITIC independent + IBM Redbook), planned-outage corrected
- Cost model: MSU-based software licensing + hardware amortization + maintenance + energy, On/Off Capacity-on-Demand (CoD) elastic peak handling
- Yearly simulation: 8,760-hour workload profiles, R4HA-aware billing, automatic permanent vs CoD strategy comparison

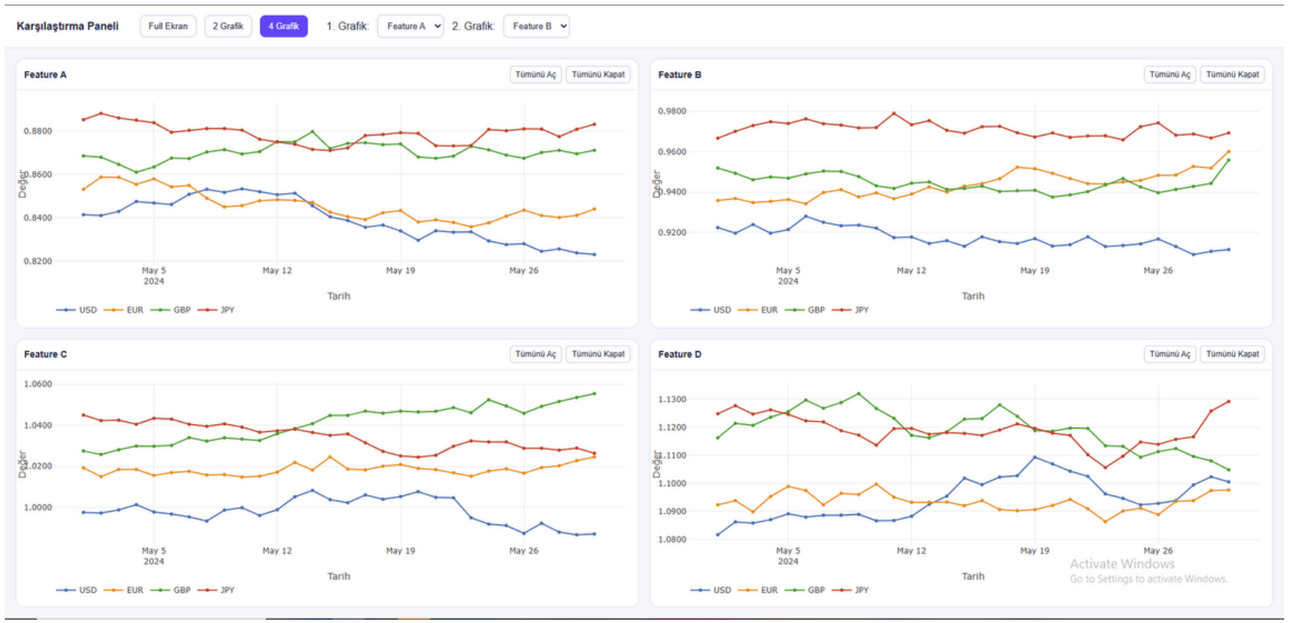
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DIFFICULTIES ENCOUNTERED

2-Interactive comparison dashboard (complete, separate tool)

- Multi-panel visualization tool for side-by-side metric comparison
- Supports 2-graph and 4-graph layouts with selectable feature pairs
- Built for visual exploration of model outputs across dimensions



Not yet implemented

- Mainframe TCO expansion: CICS/Db2 licensing, DS8900F storage, operations personnel
- Entire clustered side: EKS + Aurora PostgreSQL + Amazon MQ + EBS/S3 + operations
- Comparison layer: side-by-side output, parameter sweep, sawtooth crossover visualization
- Stochastic failure modeling

DIFFICULTIES ENCOUNTERED

1-Data opacity

- IBM does not publish mainframe hardware list prices, CICS/Db2 license fees, or GDPS premiums
- No direct TPC-C benchmark exists for mainframes

2-Scope correction mid-project

- Initial model drifted toward raw compute power/$ comparison

3-Cross-generation calibration

- MIPS is not perfectly normalized across processor generations

4-Availability measurement ambiguity

- Discovered that both IBM and ITIC report hardware-only availability (planned maintenance excluded)

TASKS TO BE COMPLETED IN THE SECOND SEMESTER

Task Breakdown and Timetable

Period	Task
Weeks 1–2	Mainframe TCO expansion: add CICS/Db2 licensing, DS8900F storage model, operations personnel costs
Weeks 3–6	Clustered model: EKS + Aurora capacity catalog, throughput bridge (HammerDB calibration), latency model (M/M/c + network), availability tiers (C1–C4), cost model, auto-scaling logic
Weeks 5–6	Clustered yearly simulation: workload profiles, reserved vs auto-scaling strategy comparison
Week 7	Comparison layer: side-by-side output, parameter sweep across TPS/availability/latency ranges
Week 8	Sawtooth visualization: crossover point identification, sensitivity analysis on key [ASSUME] parameters
Weeks 9–10	Phase 2: stochastic failure modeling (MTBF/MTTR), availability as simulated outcome
Weeks 11–12	Phase 3: Monte Carlo (1,000+ replications), cost distribution and tail-risk analysis
Weeks 13–14	Final documentation, thesis writing, presentation preparation

B-plan

- If clustered calibration data is insufficient: fall back to vendor-published benchmarks (AWS blog HammerDB numbers) with explicit [ASSUME] flags.
- If Phase 2/3 stochastic modeling proves too time-consuming: deliver Phase 1 as the core contribution (deterministic yearly TCO comparison with CoD/auto-scaling)

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Thank You

Questions & Technical Discussion

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